

DDCO LAB
Sem-III
Week: 1
Section: F
SRN: PES2UG24CS311
Name: Neeraj R Rugi

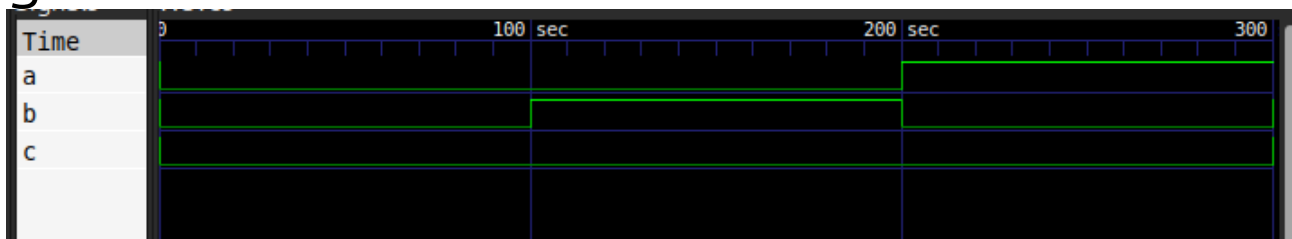
1. AND Gate:
code:

```
module    andgate(c, a, b);  
    input a;  
    input b;  
    output c;  
    assign c = a&b;  
  
endmodule
```

vvp:

```
VCD info: dumpfile andgate.vcd opened for output.  
0a = 0, b = 0, c = 0  
100a = 0, b = 1, c = 0  
200a = 1, b = 0, c = 0  
300a = 1, b = 1, c = 1
```

gtkwave:



2. OR Gate:

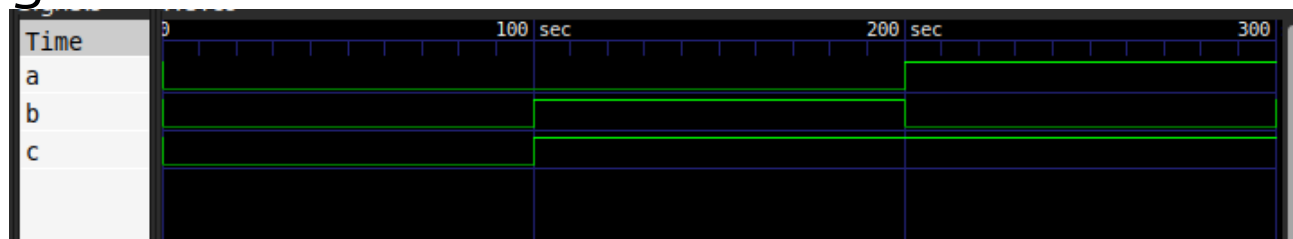
Code:

```
module    andgate(c, a, b);  
    input a;  
    input b;  
    output c;  
    assign c = a | b;  
  
endmodule
```

vvp:

```
VCD info: dumpfile orgate.vcd opened for output.  
      0a = 0, b = 0, c = 0  
     100a = 0, b = 1, c = 1  
     200a = 1, b = 0, c = 1  
     300a = 1, b = 1, c = 1
```

gtkwave:



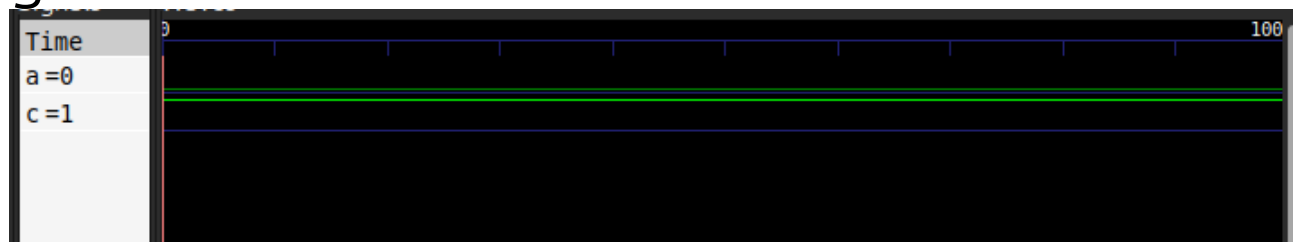
3. NOT Gate: code:

```
module notgate(c, a);  
    input a;  
    output c;  
    assign c = !a;  
endmodule
```

vvp:

```
VCD info: dumpfile notgate.vcd opened for output.  
0a = 0,c = 1  
100a = 1,c = 0
```

gtkwave:



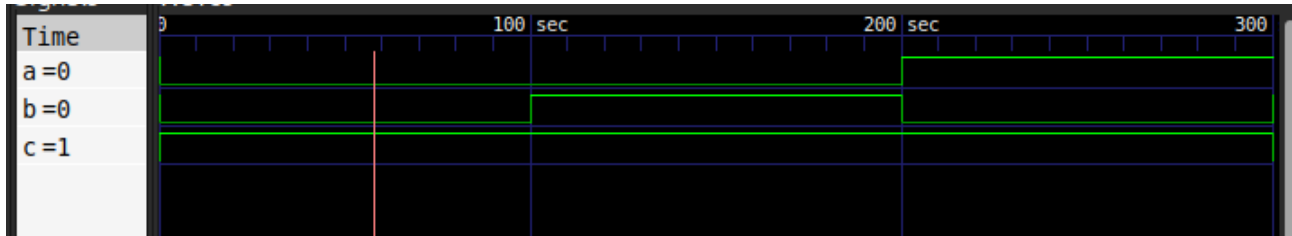
4. NAND Gate: code:

```
module    nandgate(c, a, b);  
    input a;  
    input b;  
    output c;  
    assign c = ~(a & b);  
endmodule
```

vvp:

```
VCD info: dumpfile nandgate.vcd opened for output.  
0a = 0, b = 0, c = 1  
100a = 0, b = 1, c = 1  
200a = 1, b = 0, c = 1  
300a = 1, b = 1, c = 0
```

gtkwave:



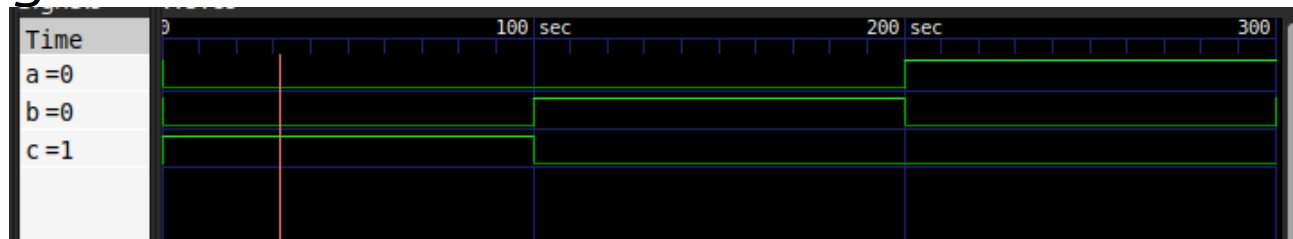
5. NOR Gate: code:

```
module norgate(c, a, b);  
    input a;  
    input b;  
    output c;  
    assign c = ~(a | b);  
endmodule
```

vvp:

```
VCD info: dumpfile norgate.vcd opened for output.  
      0a = 0, b = 0, c = 1  
     100a = 0, b = 1, c = 0  
     200a = 1, b = 0, c = 0  
     300a = 1, b = 1, c = 0
```

gtkwave:



6. XOR Gate: code:

```
module xorgate(c, a, b);  
    input a;  
    input b;  
    output c;  
    assign c = a ^ b;  
endmodule
```

vvp:

```
VCD info: dumpfile xorgate.vcd opened for output.  
    0a = 0, b = 0, c = 0  
   100a = 0, b = 1, c = 1  
   200a = 1, b = 0, c = 1  
   300a = 1, b = 1, c = 0
```

gtkwave:

